

Genetic and Clinical Characterization of Ten Unrelated Families with Autosomal-Recessive Inherited Retinal Dystrophies: An Integrated Whole-Exome and Targeted Sequencing Study

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ABSTRACT

Purpose. Inherited retinal dystrophies (IRDs) are a clinically and genetically heterogeneous group of disorders, with >300 causal genes described. Establishing a molecular diagnosis remains challenging in outbred populations and is complicated by the frequent presence of multiple rare candidate variants. We report the combined genetic and ophthalmologic characterization of ten unrelated probands with autosomal-recessive IRD, ascertained between January 2018 and August 2023 at three tertiary referral centres.

Methods. All probands underwent best-corrected visual acuity (BCVA, logMAR), full-field electroretinography (ffERG) per ISCEV 2015 standards, spectral-domain optical coherence tomography (OCT) of the central fovea, and Goldmann kinetic perimetry. Eight families (F1–F8) underwent trio whole-exome sequencing (WES) on the Illumina NovaSeq 6000 platform with the Twist Comprehensive Exome capture kit (target size 36.8 Mb). Two Ashkenazi Jewish families (F9, F10) underwent targeted Sanger sequencing of the *DHDDS* founder variant c.124A>G (p.Lys42Glu). Variants were filtered against gnomAD v3.1.2 (allele frequency <0.005) and classified by ACMG/AMP 2015 criteria.

Results. A molecular diagnosis was established in 10/10 (100%) of the families. Across F1–F8, WES yielded a mean read depth of 99.44× (SD 6.96) and a total of 664,132 variants called, of which 27 distinct candidate autosomal-recessive variants survived the variant filtering pipeline. Causal genes identified were *USH2A* (F1), *EYS* (F2), *CEP290* (F3), *PDE6B* (F4), *ABCA4* (F5), *PRPF31* (F6), *RHO* (F7), *CRB1* (F8) and *DHDDS* (F9, F10). Two families (F7 and F8) each carried an additional rare heterozygous variant in a second IRD gene (*IMPG2* and *NR2E3* respectively); however, neither satisfied autosomal-recessive segregation upon parental and sibling genotyping (see Section 3.4) and both were classified as variants of uncertain significance (VUS) per ACMG criteria.

Conclusion. Trio WES combined with population-specific targeted sequencing achieves a high diagnostic yield in AR-IRD. Multi-gene ambiguity is resolved by rigorous segregation analysis rather than by variant-level annotation alone.

Keywords. Retinitis pigmentosa; Leber congenital amaurosis; cone–rod dystrophy; whole-exome sequencing; DHDDS; Ashkenazi Jewish founder variant; ACMG classification; trio segregation analysis.

1. Introduction

Inherited retinal dystrophies (IRDs) collectively affect approximately 1 in 3,000 individuals worldwide and represent the leading cause of registrable blindness in working-age adults in industrialised countries. The clinical spectrum is broad, encompassing retinitis pigmentosa (RP), Leber congenital amaurosis (LCA), cone-rod dystrophy (CRD), Stargardt disease, and several syndromic forms. Genetic heterogeneity is striking: as of January 2025, the Retinal Information Network (RetNet) catalogues 312 IRD-associated loci, of which 287 are captured by our institutional bioinformatics pipeline (see Methods, Section 2.4). Approximately 60–70% of IRD cases follow autosomal-recessive (AR) inheritance, with the remainder distributed across autosomal-dominant, X-linked and mitochondrial modes.

Whole-exome sequencing (WES) has emerged as the workhorse for IRD gene discovery and clinical diagnosis, with reported diagnostic yields between 56% and 76% depending on cohort composition and analytical rigour. Two complicating factors recur. First, in outbred populations a proband often harbours more than one rare candidate variant in a known IRD gene; allele-frequency filtering alone is insufficient to disambiguate causality, and trio segregation analysis becomes essential. Second, in genetically isolated populations, founder alleles such as *DHDDS* c.124A>G (p.Lys42Glu) in the Ashkenazi Jewish population can be efficiently screened by targeted Sanger sequencing before more expensive panels or WES.

In this study, we describe the integrated molecular and clinical characterization of ten unrelated AR-IRD families recruited between January 2018 and August 2023. We deliberately illustrate the multi-gene ambiguity problem with two families (F7 and F8) in which a second rare candidate variant co-occurred with the true causal variant; the decisive evidence for causality came from parental and sibling genotyping rather than from in-silico prediction scores.

2. Materials and Methods

2.1 Subjects and ascertainment

Ten unrelated probands (5 male, 5 female; mean age 32.6 ± 11.6 years at last ophthalmologic examination) were enrolled after written informed consent under protocol NUMC-IRB-2017-0834 (approved 22 Aug 2017, last renewed 14 Sep 2023). Two probands self-identified as Ashkenazi Jewish (F9, F10); the remaining eight comprised five Caucasian (F1, F2, F6, F7, F8), one South Asian (F3), one African (F4) and one East Asian (F5). All probands met clinical criteria for AR inheritance, defined here as either (i) ≥ 1 similarly affected sibling with two unaffected parents, or (ii) reported parental consanguinity, or (iii) compatible pedigree pattern with no affected ancestors over three generations.

2.2 Ophthalmologic evaluation

All subjects underwent best-corrected visual acuity (BCVA) testing with the ETDRS chart at 4 m and conversion to logMAR. Full-field electroretinography (ffERG) was performed under the ISCEV 2015 standard using a Diagnosys Espion E³ system with a Ganzfeld stimulator. Scotopic responses were recorded after 30 min of dark adaptation; photopic responses after 10 min of light adaptation at a background luminance of $30 \text{ cd}\cdot\text{m}^{-2}$. Optical coherence tomography (OCT) used a Heidelberg Spectralis HRA+OCT (Heidelberg Engineering); central foveal thickness (CFT, μm) was measured automatically and confirmed manually. Goldmann kinetic perimetry was performed by experienced perimetrists using the V/4e target; the area (deg^2) was computed by planimetry.

2.3 Whole-exome sequencing (F1–F8)

Genomic DNA was extracted from peripheral blood leukocytes with the QIAamp DNA Blood Mini Kit. Library preparation used the Twist Comprehensive Exome capture kit (target size 36.8 Mb) followed by 150 bp paired-end sequencing on an Illumina NovaSeq 6000 to a mean on-target read depth $\geq 80\times$. Reads were aligned to GRCh38.p13 with BWA-MEM v0.7.17. Variant calling used the GATK v4.2.6.1 HaplotypeCaller in joint-trio mode. Variants were annotated with VEP v107 against Ensembl 107. Per-family WES quality-control metrics are reported in Table 2 (Section 3.2).

2.4 Targeted *DHDDS* sequencing (F9, F10)

Probands F9 and F10 (both Ashkenazi Jewish) underwent direct Sanger sequencing of *DHDDS* exon 3 using primers DHDDS-E3F (5'-CTGTAGCCTTAGGATCCCTGT-3') and DHDDS-E3R (5'-AAGGCTGAACAGGCATCTGT-3'). PCR conditions: 95 °C 5 min; 35 cycles of (95 °C 30 s, 58 °C 30 s, 72 °C 45 s); 72 °C 7 min. Amplicons (412 bp) were sequenced bidirectionally on an ABI 3730xl. Both probands were screened for the recurrent Ashkenazi Jewish founder variant c.124A>G (p.Lys42Glu), and segregation was confirmed in both parents and available siblings.

2.5 Variant filtering and ACMG classification

Variants were retained for analysis if they (i) had a gnomAD v3.1.2 global allele frequency < 0.005, (ii) fell within the 287-gene in-house IRD panel, (iii) were predicted protein-altering or affected canonical splice sites (± 2 bp), and (iv) showed compatible autosomal-recessive segregation (homozygous, hemizygous, or compound-heterozygous in trans). In-silico support was assessed with CADD v1.6 (Phred-scaled) and REVEL v1.3 (missense only; threshold 0.5). Variants were classified according to the ACMG/AMP 2015 framework (Richards et al., 2015) by two independent reviewers; discrepancies were resolved by consensus with a third reviewer. The diagnostic criterion for assigning a causal gene was the presence of biallelic Pathogenic or Likely Pathogenic variants segregating with disease in the family. A single heterozygous candidate, even with high CADD or REVEL, was not considered diagnostic and was reported as a VUS.

3. Results

3.1 Clinical findings

All ten probands presented with clinical and electrophysiological evidence of advanced retinal dystrophy. Mean age at symptom onset was 12.8 ± 7.6 years, and mean age at examination was 32.6 ± 11.6 years. Seven probands carried a clinical diagnosis of retinitis pigmentosa (RP), two of Leber congenital amaurosis (LCA), and one of cone-rod dystrophy (CRD). BCVA was severely reduced in both eyes (mean OD 0.91 ± 0.41 logMAR; mean OS 0.97 ± 0.39 logMAR). Full-field ERG demonstrated markedly attenuated scotopic responses in all probands (mean scotopic a-wave 16.91 ± 8.00 μ V; mean scotopic b-wave 47.92 ± 20.20 μ V), with relatively preserved photopic responses in some patients (mean photopic a-wave 18.81 ± 9.72 μ V; mean photopic b-wave 39.73 ± 16.75 μ V). Mean central foveal thickness on OCT was 161.1 ± 42.4 μ m and the mean Goldmann V/4e isopter area was 2175 ± 1073 deg². Detailed per-patient findings are given in Table 1.

Fam.	Sex	Age	Dx	Onset	BCVA OD	BCVA OS	Scot a (μ V)	Scot b (μ V)	Scot IT (ms)	Phot a (μ V)	Phot b (μ V)	Phot IT (ms)	CFT (μ m)	GVF (deg ²)
F1	M	27	RP	9	1.10	1.20	12.4	41.2	118.0	14.1	38.7	35.2	142	1820
F2	F	34	RP	14	0.80	0.90	18.1	55.3	110.0	22.4	47.9	33.1	178	2510
F3	M	19	LCA	1	1.70	1.80	3.2	10.1	132.0	4.8	12.3	38.4	98	410
F4	F	41	RP	18	0.60	0.70	22.8	61.5	108.0	26.1	52.4	32.7	195	2980
F5	M	23	CRD	11	1.00	1.05	16.3	48.7	114.0	8.2	21.6	39.1	121	1490
F6	F	52	RP	25	0.40	0.50	28.4	72.1	104.0	31.8	58.9	31.5	218	3640
F7	M	31	RP	12	0.90	0.95	14.9	44.8	116.0	19.7	43.2	34.6	156	2120
F8	F	16	LCA	2	1.40	1.30	6.7	18.9	128.0	7.1	17.4	37.9	109	720
F9	M	38	RP	16	0.70	0.75	20.6	58.4	109.0	24.3	49.8	33.4	187	2740
F10	F	45	RP	20	0.50	0.55	25.7	68.2	106.0	29.6	55.1	32.2	207	3320

Table 1. Per-patient clinical and electrophysiological findings. Dx = clinical diagnosis (RP = retinitis pigmentosa, LCA = Leber congenital amaurosis, CRD = cone-rod dystrophy); Onset = age at subjective symptom onset (years); BCVA = best-corrected visual acuity (logMAR) in right (OD) and left (OS) eye; Scot = scotopic fERG; Phot = photopic fERG; a/b = a-wave and b-wave amplitude (μ V); IT = implicit time (ms); CFT = central foveal thickness on SD-OCT (μ m); GVF = Goldmann V/4e isopter area (deg²).

3.2 Whole-exome sequencing quality and variant yield

WES was performed in eight families (F1–F8). Per-family quality metrics are summarised in Table 2. Across the eight exomes, mean read depth was $99.44\times$ (SD 6.96; range 88.4–110.6 $\times$), and the proportion of target bases covered $\geq 20\times$ was $\geq 95.2\%$ in every family. A total of 664,132 variants were called across the eight exomes; following allele-frequency, panel-membership and protein-impact filtering, 151 distinct rare variants in the 287-gene IRD panel remained as candidates, of which 27 satisfied a candidate autosomal-recessive configuration (homozygous or putative compound-heterozygous) and were carried forward to segregation analysis.

Family	Mean depth ($\times$)	% bases $\geq 20\times$	% bases $\geq 30\times$	Variants called	Rare ($<0.5\%$)	In IRD panel	Candidate AR vars
F1	102.3	97.8	94.1	84,512	421	17	2
F2	95.7	96.5	92.3	81,207	468	19	3
F3	88.4	95.2	90.1	79,884	502	22	4
F4	110.6	98.4	95.7	86,341	389	14	2
F5	98.1	96.9	93.0	82,995	447	18	3

Family	Mean depth (×)	% bases ≥20×	% bases ≥30×	Variants called	Rare (<0.5%)	In IRD panel	Candidate AR vars
F6	105.2	97.1	93.8	85,103	434	16	2
F7	101.4	97.0	93.4	83,618	455	21	5
F8	93.8	96.0	91.7	80,472	489	24	6
Total/Mean	99.44	96.86	93.01	664,132	3605	151	27

Table 2. Whole-exome sequencing QC and candidate-variant yield for the eight families undergoing trio WES (F1–F8). 'Rare (<0.5%)' refers to gnomAD v3.1.2 global allele frequency below 0.005. 'In IRD panel' counts unique variants within the 287-gene in-house panel. 'Candidate AR vars' are rare in-panel variants in a configuration compatible with autosomal recessive disease prior to segregation analysis.

3.3 Causal variants and per-family findings

Table 3 summarises the causal variants assigned to each of the ten families after full segregation analysis. A molecular diagnosis was established in 10/10 (100%) of probands. For F7 and F8, the assigned causal gene was supported by biallelic Pathogenic variants segregating with disease (see Section 3.4 for the disambiguation of the additional VUS detected in each of these two families).

Fam.	Inh.	Gene	cDNA	Protein	Zyg.	gnomAD AF	CADD	REVEL	ACMG
F1	AR	USH2A	c.2299delG	p.Glu767SerfsTer21	hom	2.3e-04	35.0	—	Pathogenic
F2	AR	EYS	c.4350_4356del	p.Ile1451PhefsTer3	comp-het	4.1e-05	33.0	—	Pathogenic
F3	AR	CEP290	c.2991+1655A>G	p.Cys998Ter	hom	1.8e-04	27.0	—	Pathogenic
F4	AR	PDE6B	c.1923_1969del	p.Glu642SerfsTer8	hom	8.0e-05	33.5	—	Pathogenic
F5	AR	ABCA4	c.5882G>A;c.6079C>T	p.Gly1961Glu;p.Leu2027Phe	comp-het	3.7e-04	28.5	0.92	Pathogenic
F6	AD	PRPF31	c.522_527+10del	p.?	het-dom*	1.0e-05	32.0	—	Pathogenic
F7	AR	RHO	c.448G>A	p.Glu150Lys	hom	5.0e-06	31.0	0.88	Pathogenic
F8	AR	CRB1	c.2843G>A;c.498_506del	p.Cys948Tyr;p.Ile167_Gly169del	comp-het	1.2e-04	29.5	0.81	Pathogenic
F9	AR	DHDDS	c.124A>G	p.Lys42Glu	hom	2.0e-03	24.0	—	Pathogenic (founder)
F10	AR	DHDDS	c.124A>G	p.Lys42Glu	hom	2.0e-03	24.0	—	Pathogenic (founder)

Table 3. Causal variants per family. Inh. = inheritance mode (AR = autosomal recessive; AD = autosomal dominant with reduced penetrance, F6 only; AR-X = recessive X-linked-equivalent presentation in F4 with biallelic *PDE6B* variants on chromosome 4). cDNA and protein nomenclature follow HGVS v21.0.0 against RefSeq mRNAs. Zyg. = zygosity in proband (hom = homozygous; comp-het = compound heterozygous in trans; het = heterozygous). gnomAD AF = global allele frequency in gnomAD v3.1.2. CADD v1.6 Phred-scaled scores; REVEL v1.3 (missense only). ACMG classification per Richards et al. 2015.

3.3.1 F1 (Caucasian, male, 27 y)

The proband was first noted to have nyctalopia at age 9 years and presented at age 24 with mid-peripheral pigmentary changes consistent with classic RP. BCVA at last examination was 1.10 logMAR OD and 1.20 logMAR OS. Scotopic ERG a- and b-wave amplitudes were 12.4 μ V and 41.2 μ V respectively (scotopic implicit time 118.0 ms). WES yielded 84,512 variants; 421 were rare and 17 fell within the IRD panel, of which 2 (both in *USH2A*) satisfied the AR filter. The proband is homozygous for *USH2A* c.2299delG (p.Glu767SerfsTer21); both unaffected parents are carriers, and trio Sanger confirmed segregation. Audiologic evaluation revealed mild high-frequency sensorineural hearing loss consistent with Usher syndrome type IIa.

3.3.2 F2 (Caucasian, female, 34 y)

Onset at age 14, with progressive ring scotoma. WES called 81,207 variants; after filtering, 19 in-panel variants remained, with 3 candidate AR configurations. The proband is compound heterozygous for *EYS* c.4350_4356del (p.Ile1451PhefsTer3) and c.8559-2A>G (splice acceptor disruption). Parental Sanger sequencing demonstrated that the c.4350_4356del allele is paternal in origin and the c.8559-2A>G allele is maternal. Both variants meet PVS1 (predicted loss-of-function) and PM2 (rare) criteria.

3.3.3 F3 (South Asian, male, 19 y)

Severe early-onset retinal dystrophy presenting at age 12 months with nystagmus, sluggish pupils and an extinguished ERG, consistent with Leber congenital amaurosis (LCA). WES called 79,884 variants. The proband is homozygous for *CEP290* c.2991+1655A>G, the deep-intronic founder variant creating a cryptic exon and bringing in a premature stop (p.Cys998Ter). Both unaffected parents are heterozygous; the family declined further testing on the only available sib.

3.3.4 F4 (African, female, 41 y)

Onset at age 18, sector-form RP progressing to diffuse involvement by age 35. The pedigree is consanguineous (parents are first cousins once removed) with one similarly affected younger sibling. WES called 86,341 variants; 14 in-panel variants survived AF filtering. The proband is homozygous for *PDE6B* c.1923_1969del (p.Glu642SerfsTer8). Both unaffected parents are heterozygous; the affected sister is also homozygous.

3.3.5 F5 (East Asian, male, 23 y)

Cone-rod dystrophy presenting at age 11 with progressive central acuity loss and bilateral macular atrophy. WES called 82,995 variants; 18 in-panel variants, with 3 AR candidates. The proband is compound heterozygous for *ABCA4* c.5882G>A (p.Gly1961Glu; paternal) and c.6079C>T (p.Leu2027Phe; maternal). REVEL scores 0.92 and 0.88 respectively; both classified Pathogenic by ACMG criteria (PM2 + PP3 + PM3_strong + PS3_supporting).

3.3.6 F6 (Caucasian, female, 52 y)

Late-onset RP with onset of symptoms at age 25 and slow progression. The pedigree shows three affected individuals across two generations (proband, mother, maternal aunt), compatible with autosomal-dominant inheritance with reduced penetrance. WES called 85,103 variants; in-panel filtering identified a heterozygous splicing variant in *PRPF31*: c.522_527+10del. The variant is paternally inherited from an unaffected obligate carrier (consistent with the known incomplete penetrance of *PRPF31* mutations). Although the inheritance is technically autosomal-dominant rather than recessive, F6 was included in this study because the diagnostic pipeline was the same.

3.4 Multi-gene candidate ambiguity in families F7 and F8

Two of the eight WES-sequenced families (F7 and F8) carried, in addition to a clearly biallelic causal variant in a canonical IRD gene, a second rare heterozygous variant in another IRD-panel gene. Such co-occurrence is not uncommon in outbred populations and is a well-known source of diagnostic error if segregation analysis is not rigorously applied. Below we describe each family in detail and summarise the additional candidate variants in Table 4.

3.4.1 F7 (Caucasian, male, 31 y) — *RHO* vs *IMPG2*

F7 presented with classical RP, onset at age 12, with progressive nyctalopia, ring scotoma, bone-spicule pigmentation and an attenuated scotopic ERG (a-wave 14.9 μ V, b-wave 44.8 μ V, implicit time 116.0 ms). WES called 83,618 variants; 21 in-panel variants survived AF filtering, of which 5 were configured as candidate AR. Two variants stood out as plausible:

- *RHO* c.448G>A (p.Glu150Lys): present in the proband in the **homozygous** state. gnomAD AF = 5.0×10^{-5} (one heterozygote in 251,496 alleles). CADD 31.0; REVEL 0.88. Both unaffected parents are heterozygous; unaffected sibling 1 is heterozygous and clinically and electrophysiologically normal; unaffected sibling 2 is wild-type. Although *RHO* variants are most commonly autosomal-dominant, a small number of biallelic *RHO* recessive RP cases have been reported (Rosenfeld et al., 1992; Kumaramanickavel et al., 1994), in which heterozygous carriers are asymptomatic. The parental asymptomatic carrier status here is therefore consistent with recessive *RHO* RP.
- *IMPG2* c.3257G>A (p.Arg1086His): present in the proband in the **heterozygous** state only. gnomAD AF = 1.4×10^{-3} (approximately 350 heterozygotes in gnomAD v3.1.2). CADD 23.5; REVEL 0.41. Segregation analysis: the variant is present in the unaffected father (heterozygous) and absent in the unaffected mother. No second *IMPG2* rare allele was identified in the proband on manual review of CNV calls or splice-region sequence. The *IMPG2* variant is therefore present only as a single heterozygous allele in F7, does not segregate biallelically, and cannot account for a recessive disease. We classify *IMPG2* c.3257G>A as a variant of uncertain significance (VUS) and assign ***RHO* c.448G>A homozygous as the causal variant** for family F7.

3.4.2 F8 (Caucasian, female, 16 y) — *CRB1* vs *NR2E3*

F8 was diagnosed with LCA at age 2 years on the basis of severe early-onset visual impairment, nystagmus, and an extinguished ERG (scotopic a 6.7 μ V, b 18.9 μ V; photopic a 7.1 μ V, b 17.4 μ V). WES called 80,472 variants; 24 in-panel variants and 6 AR-candidate configurations. Two distinct candidate genes were identified:

- *CRB1* c.2843G>A (p.Cys948Tyr) and c.498_506del (p.Ile167_Gly169del): the proband is **compound heterozygous** for these two variants. Parental Sanger sequencing demonstrates that c.2843G>A is carried by the unaffected father, and c.498_506del by the unaffected mother — therefore the two variants are in trans. Both variants are previously reported as Pathogenic in *CRB1*-LCA (Lotery et al., 2001; den Hollander et al., 2004). gnomAD AF = 1.2×10^{-5} and 4.5×10^{-5} , respectively. CADD 29.5; REVEL 0.81.
- *NR2E3* c.119-2A>C (canonical splice acceptor): the proband is heterozygous; the unaffected father is heterozygous; one unaffected sibling is also heterozygous; the unaffected mother is wild-type. The variant occurs at gnomAD AF = 2.1×10^{-3} (approximately 530 heterozygotes), with no homozygotes reported. As the proband carries only a single heterozygous allele, and as the variant is also present heterozygously in an unaffected sibling, the *NR2E3* finding does not satisfy autosomal-recessive segregation. We classify *NR2E3* c.119-2A>C as a variant of uncertain significance (VUS) and assign **biallelic *CRB1* variants (c.2843G>A and c.498_506del in trans) as the causal genotype** for family F8.

Family	Gene	cDNA	Protein	Zyg.	gnomAD AF	CADD	REVEL	ACMG	Disposition
F7	IMPG2	c.3257G>A	p.Arg1086His	het	1.4e-03	23.5	0.41	VUS	Single heterozygous allele; mother wild-type. Does not satisfy
F8	NR2E3	c.119-2A>C	p.?	het	2.1e-03	22.0	—	VUS	Single heterozygous allele; also carried by unaffected sib. Does not satisfy

Table 4. Additional rare candidate variants identified in F7 and F8 that did NOT meet the diagnostic criteria for causality. Both variants were classified as VUS and reported in the clinical diagnostic report as incidental findings only. The disposition column summarises the segregation rationale for excluding them from diagnostic causality.

3.4.3 Segregation summary across all ten families

Trio (proband + both parents) genotyping was performed by Sanger sequencing of the candidate variants in every family. The full segregation pattern is summarised below.

F1: Proband hom c.2299delG; both unaffected parents het; one unaffected sib het, one unaffected sib wt.

F2: Proband comp-het c.4350_4356del / c.8559-2A>G; father carries c.4350_4356del, mother carries c.8559-2A>G; two unaffected sibs each carry one allele.

F3: Proband hom c.2991+1655A>G; both unaffected parents het; no sibs.

F4: Proband hom c.1923_1969del; both unaffected parents het; one affected sib also hom (consanguineous).

F5: Proband comp-het c.5882G>A / c.6079C>T; father het c.5882G>A, mother het c.6079C>T; one unaffected sib wt.

F6: Proband het c.522_527+10del (PRPF31, autosomal dominant with reduced penetrance); affected mother carries same allele.

F7: Proband carries (a) RHO c.448G>A homozygous AND (b) IMPG2 c.3257G>A heterozygous. Both unaffected parents are heterozygous for RHO c.448G>A. Unaffected sib 1 is heterozygous for RHO c.448G>A and wild-type for IMPG2. Unaffected sib 2 is wild-type for both. IMPG2 c.3257G>A is present heterozygously in the unaffected father only; the mother is wild-type for IMPG2.

F8: Proband carries (a) CRB1 c.2843G>A heterozygous AND CRB1 c.498_506del heterozygous (in trans, parental phasing) AND (b) NR2E3 c.119-2A>C heterozygous. Father carries CRB1 c.2843G>A only; mother carries CRB1 c.498_506del only. NR2E3 c.119-2A>C is also carried heterozygously by the unaffected father and one unaffected sib, and homozygously by NO family member. NR2E3 segregation therefore does NOT support recessive causality.

F9: Proband hom DHDDS c.124A>G; both unaffected parents het; one unaffected sib het.

F10: Proband hom DHDDS c.124A>G; both unaffected parents het; no sibs.

3.3.9 F9 (Ashkenazi Jewish, male, 38 y)

Classical RP with onset at age 16. The proband was screened by targeted Sanger sequencing for the Ashkenazi Jewish founder variant *DHDDS* c.124A>G (p.Lys42Glu). The proband is homozygous; both unaffected parents are heterozygous; one unaffected sib is heterozygous. The variant carrier frequency in the Ashkenazi Jewish population is approximately 1 in 322 (gnomAD v3.1.2 AJ-specific AF = 2.0×10^{-3}).

3.3.10 F10 (Ashkenazi Jewish, female, 45 y)

Late-stage RP with onset at age 20. As with F9, targeted Sanger sequencing of *DHDDS* exon 3 identified homozygous c.124A>G (p.Lys42Glu). Both unaffected parents are heterozygous. No siblings were available for testing.

3.5 Cohort summary statistics

Table 5 reports the mean $\pm$ standard deviation of the principal clinical and electrophysiological measurements across the full ten-family cohort. All values are derived directly from the per-patient data in Table 1.

Measurement	Mean	SD	Min	Max
Age at exam (y)	32.60	11.64	16.00	52.00
Age at onset (y)	12.80	7.55	1.00	25.00
BCVA OD (logMAR)	0.91	0.41	0.40	1.70
BCVA OS (logMAR)	0.97	0.39	0.50	1.80
Scotopic a-wave (μ V)	16.91	8.00	3.20	28.40
Scotopic b-wave (μ V)	47.92	20.20	10.10	72.10
Photopic a-wave (μ V)	18.81	9.72	4.80	31.80
Photopic b-wave (μ V)	39.73	16.75	12.30	58.90
OCT central foveal thickness (μ m)	161.10	42.35	98.00	218.00
Goldmann V/4e isopter area (deg ²)	2175.00	1073.27	410.00	3640.00

Table 5. Summary statistics across the ten-family cohort. Means and standard deviations are computed across all 10 probands. Min/Max give the cohort range.

4. Discussion

We report a high diagnostic yield (10/10, 100%) in this small but deeply phenotyped cohort of ten unrelated AR-IRD families. The yield exceeds the 56–76% range typically reported for unselected WES cohorts and is attributable to the rigorous clinical preselection (advanced bilateral retinal dystrophy with compatible AR pedigree) and to the use of population-specific founder-allele screening (*DHDDS* p.Lys42Glu) in the two Ashkenazi Jewish probands prior to consideration of WES.

The most diagnostically instructive families in our series are F7 and F8. In each of these two probands, allele-frequency filtering and in-silico prediction alone would have permitted either of two IRD-panel genes as causal — *RHO* vs *IMPG2* in F7, and *CRB1* vs *NR2E3* in F8. Only careful trio segregation analysis distinguished the truly biallelic causal genotype from the incidental heterozygous VUS. Specifically, in F7 the *IMPG2* candidate appeared in a single heterozygous configuration in both the proband and the unaffected father, with no second allele identifiable on the maternal haplotype; in F8 the *NR2E3* candidate was carried heterozygously by an unaffected sibling, thereby failing to track with affection status. Diagnostic pipelines that do not enforce parental and sibling genotyping risk reporting the wrong gene in approximately 20% of outbred IRD cohorts on current estimates (Carss et al., 2017; Stone et al., 2017).

The *DHDDS* p.Lys42Glu founder variant remains the single most common AR-RP allele in the Ashkenazi Jewish population, with a carrier frequency of approximately 1 in 322 in our institutional registry and a comparable frequency in gnomAD v3.1.2 (AJ-specific AF 2.0×10^{-3}). Targeted Sanger sequencing of *DHDDS* exon 3 is therefore a cost-effective first-tier test in this population and has a >25% diagnostic yield in AJ probands with isolated RP (Zelinger et al., 2011; Züchner et al., 2011).

Limitations of the present study include the modest cohort size, the single-centre design, and the absence of long-read or optical-mapping data to exclude structural variants in the remaining unsolved candidate-AR loci in F7 and F8 (Table 4). Functional validation of the biallelic *CRB1* genotype in F8 by minigene or retinal-organoid assays is in progress.

5. Conclusions

Combined trio WES and population-specific targeted sequencing achieves a 100% molecular diagnostic yield in this AR-IRD cohort. Multi-gene candidate ambiguity, present in 2/8 (25%) of the WES families, was resolved exclusively by parental and sibling segregation analysis. We highlight *RHO* c.448G>A (p.Glu150Lys) homozygous in F7 and biallelic *CRB1* in F8 as the correct diagnostic assignments, with the co-occurring *IMPG2* and *NR2E3* heterozygous variants classified as incidental VUS.

Supplementary Materials

Table S1. Allele-count breakdown in gnomAD v3.1.2 for each causal variant reported in Table 3.

Gene	cDNA	Tot. alleles	Tot. hom	AF global	AF AFR	AF EAS	AF NFE	AF SAS	AF AJ
USH2A	c.2299delG	252,012	0	2.3e-04	1.0e-04	0.0e+00	3.1e-04	1.5e-04	0.0e+00
EYS	c.4350_4356del	251,960	0	4.1e-05	0.0e+00	0.0e+00	6.3e-05	0.0e+00	0.0e+00
CEP290	c.2991+1655A>G	251,488	0	1.8e-04	2.6e-05	0.0e+00	2.5e-04	4.4e-04	0.0e+00
PDE6B	c.1923_1969del	251,774	0	8.0e-05	1.2e-04	0.0e+00	1.0e-04	0.0e+00	0.0e+00
ABCA4	c.5882G>A	251,402	1	3.7e-04	5.2e-05	1.0e-04	5.8e-04	1.3e-04	0.0e+00
ABCA4	c.6079C>T	251,488	0	4.5e-05	0.0e+00	0.0e+00	6.9e-05	0.0e+00	0.0e+00
PRPF31	c.522_527+10del	251,498	0	1.0e-05	0.0e+00	0.0e+00	1.5e-05	0.0e+00	0.0e+00
RHO	c.448G>A	251,496	0	5.0e-06	0.0e+00	0.0e+00	9.6e-06	0.0e+00	0.0e+00
CRB1	c.2843G>A	251,492	0	1.2e-04	4.0e-05	0.0e+00	1.8e-04	5.1e-05	0.0e+00
CRB1	c.498_506del	251,488	0	4.5e-05	0.0e+00	0.0e+00	7.0e-05	0.0e+00	0.0e+00
DHDDS	c.124A>G	251,496	0	7.4e-05	0.0e+00	0.0e+00	8.2e-06	0.0e+00	2.0e-03

AFR = African; EAS = East Asian; NFE = Non-Finnish European; SAS = South Asian; AJ = Ashkenazi Jewish (gnomAD v3.1.2 population labels).

Table S2. Longitudinal BCVA progression (logMAR, OD) across three consecutive annual visits, V1 → V3, for each proband.

Family	V1 (Year -2)	V2 (Year -1)	V3 (current)	Δ V1→V3
F1	0.98	1.05	1.10	+0.12
F2	0.68	0.75	0.80	+0.12
F3	1.58	1.65	1.70	+0.12
F4	0.48	0.55	0.60	+0.12
F5	0.88	0.95	1.00	+0.12
F6	0.28	0.35	0.40	+0.12
F7	0.78	0.85	0.90	+0.12
F8	1.28	1.35	1.40	+0.12
F9	0.58	0.65	0.70	+0.12
F10	0.38	0.45	0.50	+0.12

All values in logMAR units, right eye (OD). Δ V1→V3 = mean annual progression × 2 years.

Table S3. Goldmann V/4e isopter area (deg²) decomposed by quadrant (SN = superonasal, SI = superotemporal-inferior etc.).

Family	SN	ST	IN	IT	Total
F1	459	431	433	497	1820

Family	SN	ST	IN	IT	Total
F2	599	632	611	668	2510
F3	105	94	102	109	410
F4	750	789	766	675	2980
F5	389	346	379	376	1490
F6	819	968	888	965	3640
F7	564	514	497	545	2120
F8	161	173	202	184	720
F9	740	729	642	629	2740
F10	848	755	889	828	3320

Quadrant areas computed by planimetry on each individual visual-field diagram and summed.

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